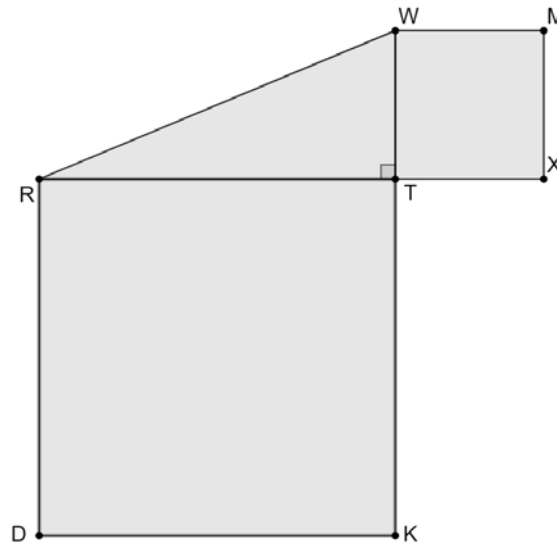


A diagram is shown where $RT = 12$ and $WT = 5$.



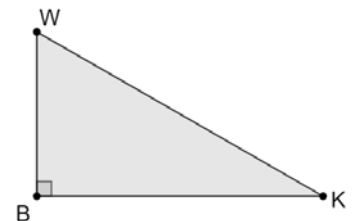
Complete the statements.

1. To show the Pythagorean theorem, a square that is drawn on the hypotenuse of $\triangle TWR$ will have an area equal to:

- ☐ $WT + RT$.
- ☐ $(WT + RT)^2$.
- ☐ the sum of the area of square $WMXT$ and the area of square $RTKD$.

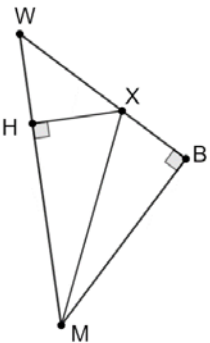
2. The length of each side of the square that is drawn with side $\overline{RW}$ is _____.

3. $\triangle WKB$ is shown where $WK = \sqrt{65}$ and $WB = 4$. Find the exact length of $\overline{BK}$.

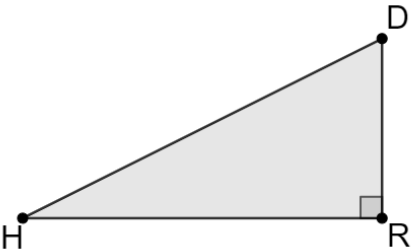


4. A right triangle has side lengths of $\sqrt{35}$ and $\sqrt{52}$. Find the exact length of the hypotenuse.

5. Triangle WBM is shown where $m\angle WBM = 90^\circ$ and $m\angle XHM = 90^\circ$.
 $MX = \sqrt{29}$, $WX = 3$, and $BM = 5$.
 Determine the length of $\overline{WM}$. If necessary, round your answer to the nearest thousandth.

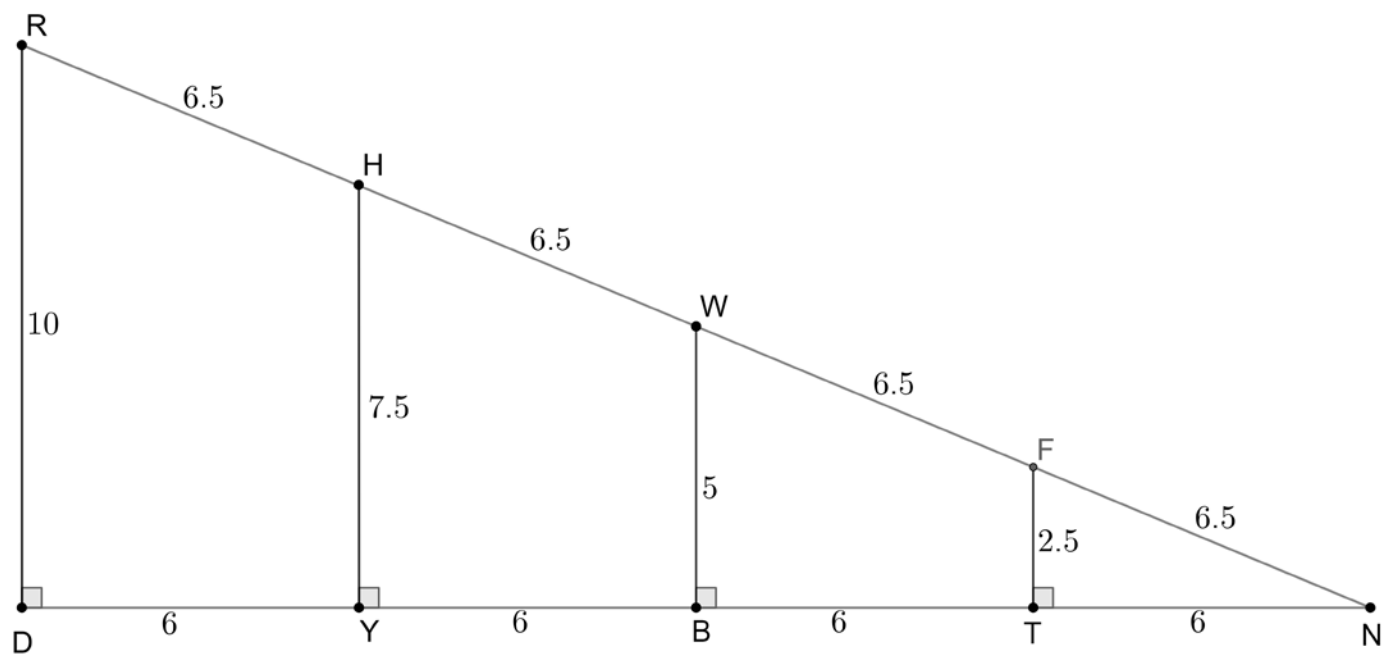


$\triangle DRH$ is shown. Complete the table by finding the missing lengths. Round your answer to the nearest thousandth. Show your work below the table.



	Length of $\overline{DR}$	Length of $\overline{HR}$	Length of $\overline{DH}$
6.	28	15	
7.	36		85
8.		$\sqrt{18}$	18
9.	33		65
10.		$\sqrt{7}$	$\sqrt{32}$

The diagram shows measurements for $\triangle RND$, $\triangle HNY$, $\triangle WNB$, and $\triangle FNT$.



Complete the table using $\angle N$ as the reference angle.

	Triangle	$\frac{\text{opposite}}{\text{hypotenuse}}$	$\frac{\text{adjacent}}{\text{hypotenuse}}$	$\frac{\text{opposite}}{\text{adjacent}}$
1.	$\triangle FNT$			
2.	$\triangle WNB$			
3.	$\triangle HNY$			
4.	$\triangle RND$			

For $\triangle FNT$, $\triangle WNB$, $\triangle HNY$, and $\triangle RND$ find the decimal equivalent for each ratio.

	Ratio	$\triangle FNT$	$\triangle WNB$	$\triangle HNY$	$\triangle RND$
5.	$\frac{\text{opposite}}{\text{hypotenuse}}$				
6.	$\frac{\text{adjacent}}{\text{hypotenuse}}$				
7.	$\frac{\text{opposite}}{\text{adjacent}}$				

8. Complete the statements.

The decimal equivalents of the $\frac{\text{opposite}}{\text{hypotenuse}}$ ratios for triangles FNT , WNB , HNY , and

RND are ☐ equal.
☐ not equal.

The decimal equivalents of the $\frac{\text{adjacent}}{\text{hypotenuse}}$ ratios for

triangles FNT , WNB , HNY , and RND are ☐ equal.
☐ not equal.

of the $\frac{\text{opposite}}{\text{adjacent}}$ ratios for triangles FNT , WNB , HNY , and RND are

☐ equal.
☐ not equal.

9. Complete the statement.

Since triangles FNT , WNB , HNY , and RND are

☐ acute
☐ obtuse
☐ right

triangles that share

☐ $\angle N$
☐ side $\overline{DN}$
☐ sides $\overline{DN}$ and $\overline{RN}$

the triangles are

☐ congruent.
☐ similar.

10. Complete the statement.

Because triangles FNT , WNB , HNY , and RND are

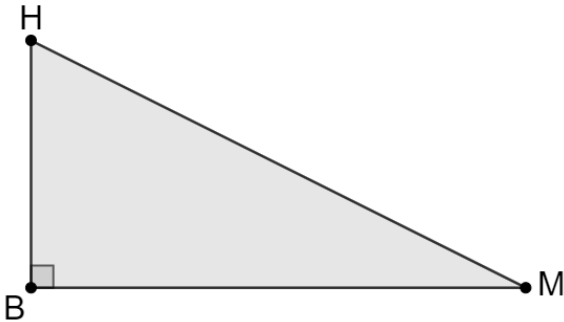
☐ congruent
☐ similar

their side lengths

are

☐ congruent.
☐ proportional.

Triangle HBM is shown where $m\angle B = 90^\circ$.



Complete the table using the given information.

	<i>HB</i>	<i>BM</i>	<i>HM</i>	Reference Angle	Cosine	Sine	Tangent
1.	8	15		$\angle M$			
2.	8	15		$\angle H$			
3.		56	65	$\angle M$			
4.	$\sqrt{23}$		$2\sqrt{10}$	$\angle M$			
5.	$\sqrt{23}$		$2\sqrt{10}$	$\angle H$			

6.

$5\sqrt{2}$

$3\sqrt{7}$

$\angle H$

7.

$2\sqrt{119}$

24

$\angle H$

8.

$\angle M$

$\frac{45}{53}$

$\frac{28}{53}$

9.

$\angle M$

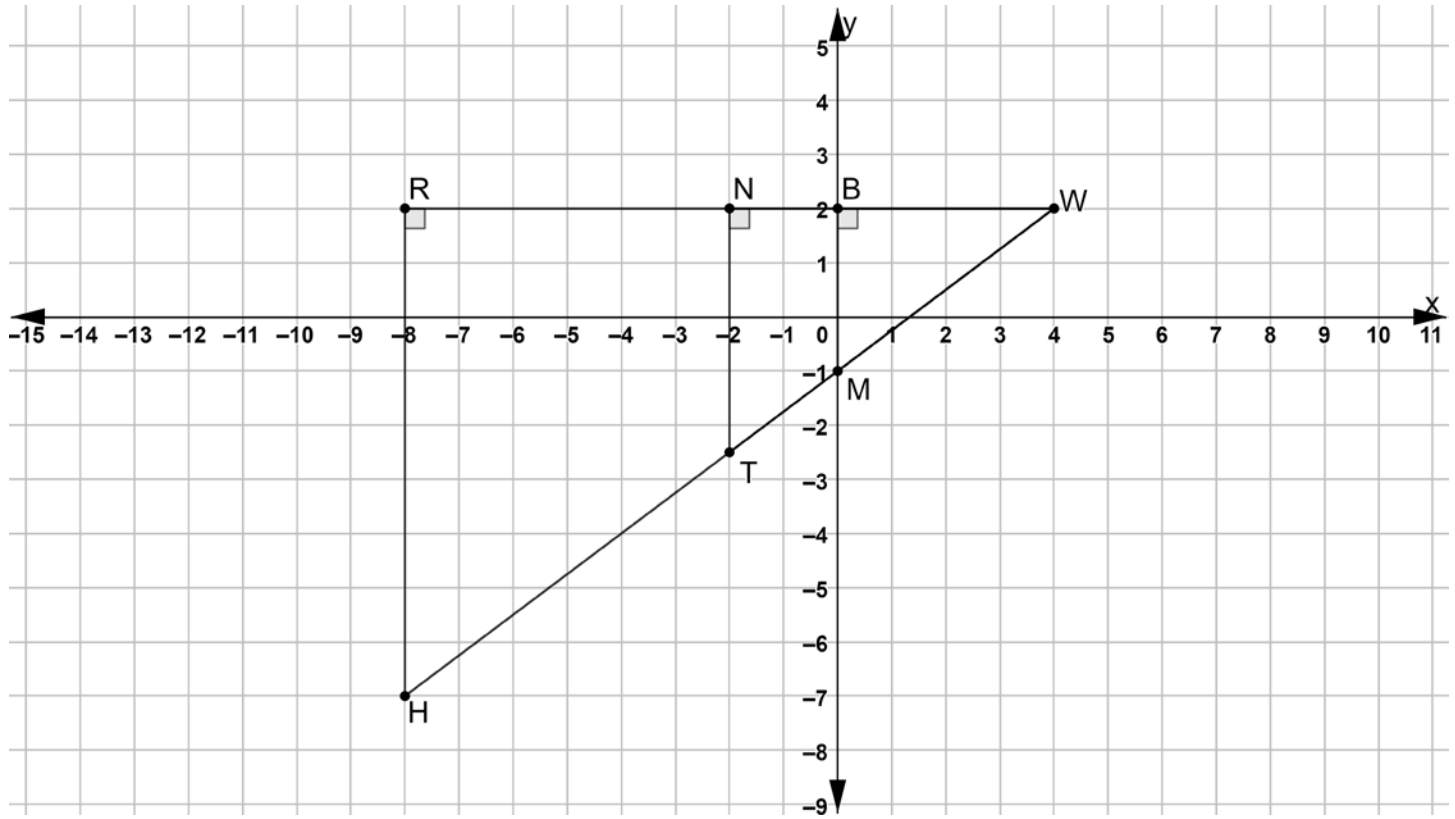
$\frac{6}{8}$

10

$\angle H$

$\frac{12}{5}$

Three right triangles are shown on the coordinate grid.



The coordinates of the vertices are shown.

$R(-8, 2)$

$N(-2, 2)$

$B(0, 2)$

$W(4, 2)$

$H(-8, -7)$

$T(-2, -2.5)$

$M(0, -1)$

1. Explain why these triangles are similar.

Complete the table.

	Triangle	Side	Length
2.	BWM	$\overline{BW}$	
		$\overline{BM}$	
		$\overline{MW}$	
3.	NWT	$\overline{NW}$	
		$\overline{TN}$	
		$\overline{TW}$	
4.	RWH	$\overline{RW}$	
		$\overline{RH}$	
		$\overline{HW}$	

Using $\angle W$ as the reference angle, find the three ratios for each triangle. Write the values as a decimal rounded to the nearest thousandth.

	Triangle	Cosine	Sine	Tangent
5.	BWM			
6.	NWT			
7.	RWH			

Use your calculator to find each. Round to the nearest thousandth.

	Trigonometric Ratio	Value rounded to nearest thousandth
8.	$\cos(37^\circ)$	
9.	$\sin(37^\circ)$	
10.	$\tan(37^\circ)$	

Geometry

Unit 7

Lesson 5 Homework

Name _____

Date _____

Period _____

$\triangle YRM$ is a right triangle where $m\angle YRM = 90^\circ$, $RM = 8$, and $YM = 17$.

1. Determine $m\angle YMR$ using a sine ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$\triangle WHN$ is a right triangle where $m\angle NWH = 90^\circ$, $NW = 48$, and $WH = 55$.

2. Determine $m\angle WNH$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$\triangle XPM$ is a right triangle where $m\angle XPM = 90^\circ$, $XP = 16$, and $XM = 20$.

3. Determine the length of $\overline{MP}$. If necessary, round your answer to the nearest thousandth.

4. Determine $m\angle PMX$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$\triangle LBZ$ is a right triangle where $m\angle BZL = 90^\circ$, $BZ = 24$, and $LB = 51$.

5. Determine the length of $\overline{ZL}$. If necessary, round your answer to the nearest thousandth.

6. Determine $m\angle LBZ$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

$\triangle MDH$ is a right triangle where $m\angle DHM = 90^\circ$, $m\angle DMH = 48^\circ$, and $HM = 14$.

7. Determine the length of $\overline{DH}$ using a tangent ratio. Show your work. If necessary, round your answer to the nearest thousandth.

8. Explain why neither the cosine nor the sine ratio could be used to determine the length of $\overline{DH}$.

$\triangle TLS$ is a right triangle where $m\angle LTS = 90^\circ$, $m\angle TSL = 63^\circ$, and $LS = 63$.

9. Determine the length of $\overline{TL}$. Show your work. If necessary, round your answer to the nearest thousandth.

10. Determine the length of $\overline{TS}$. Show your work. If necessary, round your answer to the nearest thousandth.

Geometry

Unit 7

Lesson 6 Homework

Name _____

Date _____

Period _____

$\triangle YWT$ is a right triangle where $m\angle YTW = 90^\circ$, $m\angle YWT = 30^\circ$, and $YT = 7$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

1.

2.

Side	Length
$\overline{WT}$	
$\overline{YW}$	

$\triangle RXH$ is a right triangle where $m\angle XRH = 90^\circ$, $m\angle RXH = 45^\circ$, and $XR = 12$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

3.

4.

Side	Length
$\overline{RH}$	
$\overline{XH}$	

$\triangle KGM$ is a right triangle where $m\angle MGK = 90^\circ$, $m\angle GKM = 60^\circ$, and $KM = 16$. Determine the length of each side. Show your work. Leave your answer as an exact answer.

5.

6.

Side	Length
$\overline{KG}$	
$\overline{GM}$	

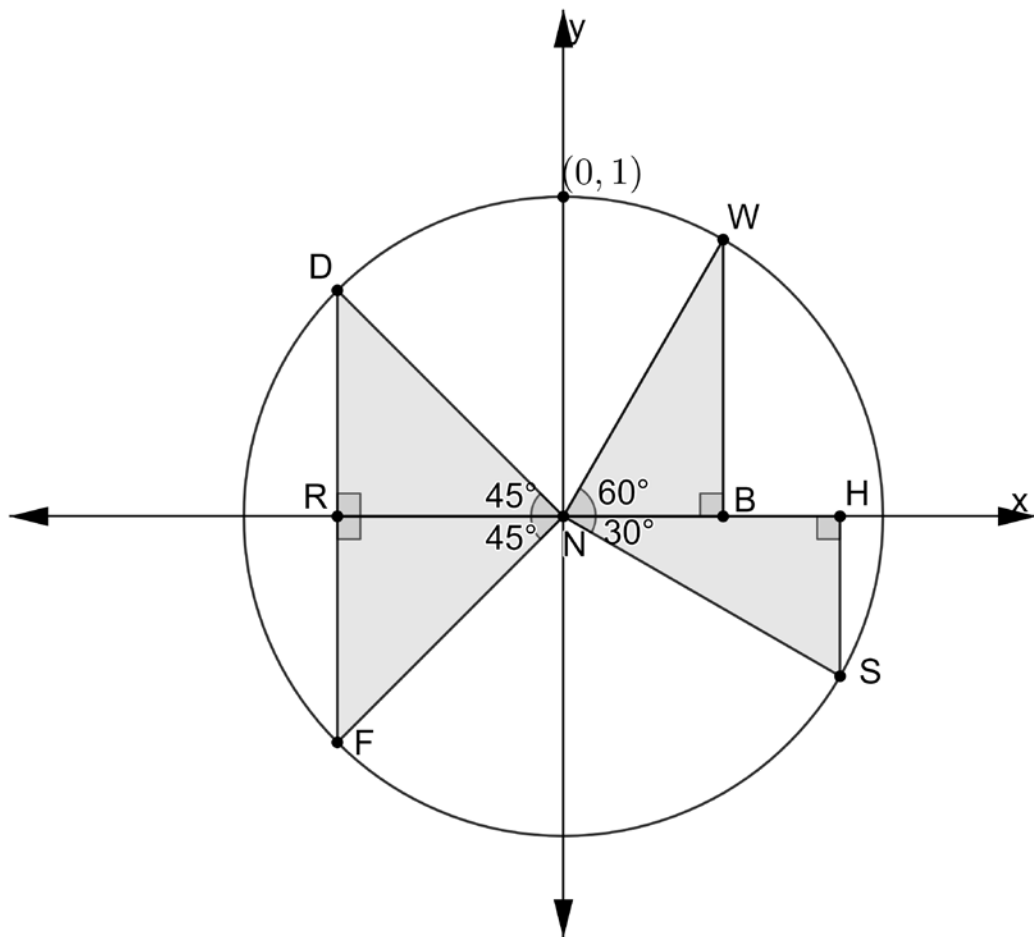
$\triangle FND$ is a right triangle where $m\angle FND = 90^\circ$, $m\angle NDF = 45^\circ$, and $DF = 36$. Determine the length of each side. Show your work. If necessary, round your answer to the nearest thousandth.

	Side	Length
7.	$\overline{ND}$	
8.	$\overline{NF}$	

$\triangle YBM$ is a right triangle where $m\angle YMB = 90^\circ$, $m\angle MYB = 30^\circ$, and $YB = 24$. Determine the length of each side. Show your work. If necessary, round your answer to the nearest thousandth.

	Side	Length
9.	$\overline{MB}$	
10.	$\overline{YM}$	

Circle N and right triangles NWB , NHS , DNR , and RFN are shown on the coordinate plane.



1. What is the length of the radius of circle N ?

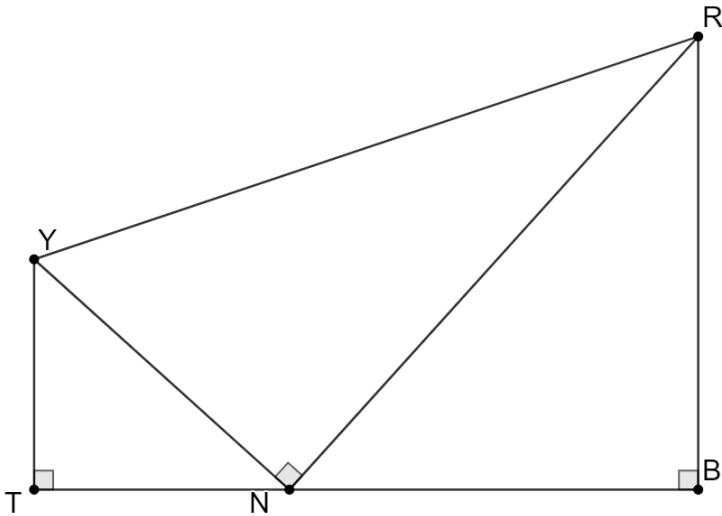
Complete the table.

	Segment	Length
2.	$\overline{NW}$	
3.	$\overline{DN}$	

Complete the table.

	Point	Coordinates
4.	B	
5.	H	
6.	R	
7.	W	
8.	D	
9.	F	
10.	S	

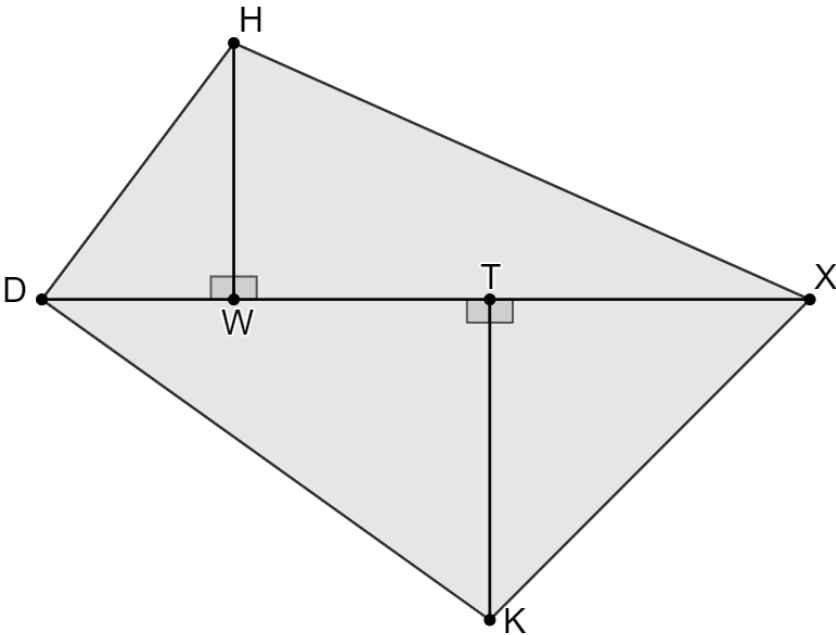
Right triangles YTN , RNY , and NRB are shown where $BR = 6$, $m\angle BNR = 48^\circ$ and $m\angle NRY = 29^\circ$.



Complete the table. In the work column, write the equation or trigonometric ratio used to determine the side measure.

	Side	Work	Side Measure
1.	$\overline{NB}$		
2.	$\overline{NR}$		
3.	$\overline{YR}$		
4.	$\overline{YN}$		
5.	$\overline{YT}$		

Quadrilateral $HXKD$ is composed of right triangles DHW , WHX , TXK , and DTK . $WH = 4$, $TK = 5$, $m\angle WDH = 53^\circ$, $m\angle WHX = 66^\circ$, $m\angle TKX = 45^\circ$, and $m\angle TDK = 35^\circ$.



Complete the table. In the work column, write the equation or trig ratio used to determine the side measure

	Side	Work	Side Measure
6.	$\overline{HX}$		
7.	$\overline{XK}$		
8.	$\overline{DK}$		
9.	$\overline{HD}$		

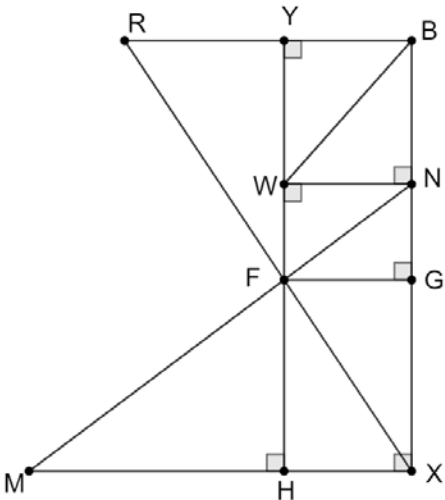
10. What is the perimeter of quadrilateral $HXKD$?

A pastry chef is creating a sculpture using chocolate. To ensure the sculpture has a strong base the chef uses a computer program to outline the structure he will use. The outline of the structure is shown.

The table shows known measurements of the structure.

$m\angle FMH = 37^\circ$	$FH = 4$ feet	$GN = 2$ feet
$m\angle XFH = 34^\circ$	$BN = 3$ feet	$m\angle YRF = 57^\circ$

Complete the table. In the work column, write the equation or trigonometric ratio used to determine the side measure.



	Side	Work	Side Measure
1.	$\overline{HX}$		
2.	$\overline{FM}$		
3.	$\overline{RB}$		

4. The polygon $FMXBR$ is the base of the chocolate structure. What is the area of the base?

The diagram shows a staircase that consists of 6 steps. The length of each step is 11 inches (in.) and the height each step is 5.25 in.

5. Determine the measure of $\angle CLR$. Round to the nearest hundredth.

6. Determine the length of $\overline{LW}$. Round to the nearest hundredth.

7. Determine the measure of $\angle HWT$. Round to the nearest hundredth.

8. The homeowner plans to paint quadrilateral $HLNT$ using chalkboard paint. What is the area of quadrilateral $HLNT$? Round to the nearest hundredth.

The crown of an oak tree is the area that consists of the branches and leaves of the tree. Leigh Ann was trying to determine the height of an oak tree's crown, TF . The height to Leigh Ann's eye, PW , is 5.5 feet (ft). Leigh Ann is standing 9 feet away from the tree. The angle of elevation from Leigh Ann's eye to the bottom of the oak tree's crown is 34° . The angle of elevation from Leigh Ann's eye to the top of the oak tree's crown is 45° .

9. Determine the height, in feet, of the tree. Round to the nearest tenth.

10. Determine the height, in feet, of the oak tree's crown. Round to the nearest tenth.

